

Computational Approach of Signaling Games

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1 Preliminaries

Signaling game

A signaling game is defined by :

- a finite set of types T ,
- a finite set of signals S ,
- a finite set of actions A ,
- payoff functions $u_S : T \times S \times A \rightarrow \mathbb{R}$ and $u_R : T \times S \times A \rightarrow \mathbb{R}$.

There is also a prior $p \in \Delta(T)$.

Strategy profiles of the Signaling game

Let M be a set of cheap talk messages. A strategy for the sender is $\sigma_S : T \rightarrow \Delta(S \times M)$, and a strategy for the receiver is a $\sigma_R : S \times M \rightarrow \Delta(A)$.

The interim payoff v_t of the sender of type $t \in T$ in the strategy profile is defined as

$$v_t = \sum_{\substack{(s,m) \in S \times M \\ a \in A}} \sigma_S(s, m|t) \sigma_R(a|s, m) u_S(t, s, a), \quad \text{for every } t \in T,$$

BNE of the Signaling game

A strategy profile is a BNE iff

- Sender optimality

For all $t \in T$ and all alternative signal and message s', m' ,

$$\sum_{(s,m) \in S \times M} \sum_{a \in A} \sigma_S(s, m | t) \sigma_R(a | s, m) U_S(t, s, a) \geq \sum_{a \in A} \sigma_R(a | s', m') U_S(t, s', a).$$

- Receiver optimality.

For all $(s, m) \in S \times M$ and all alternative actions a' ,

$$\sum_{t \in T} \sum_{a \in A} p(t) \sigma_S(s, m | t) \sigma_R(a | s, m) U_R(t, s, a) \geq \sum_{t \in T} p(t) \sigma_S(s, m | t) U_R(t, s, a').$$

Outcome of a BNE

The outcome π of a BNE is an element of $\Delta(S \times A)^T$.

With $\pi(s, a|t) = \sum_m \sigma_S(s, m | t) \sigma_R(a | s, m)$.

We note all the outcomes of all the BNE with prior p as $\mathcal{O}^{BNE}(G, p)$.

Theorem

$v \in \mathbb{R}^T$ is an interim BNE payoff of the sender in the signaling game with prior p_0 iff $v \in \text{IR}$ and $(v, p_0) \in \text{co}_{\text{flat}}(G)$.

with :

- $G = \left\{ (v, p) \in \mathbb{R}^{|T|} \times \Delta(T) : \exists s \in S, \exists y \in Y(s, p), \text{ such as } \forall t, v_t \geq u_S(t, s, y) \text{ and } v_t = u_S(t, s, y) \text{ if } p(t) > 0 \right\}.$
- $\text{co}_{\text{flat}}(G) := \left\{ (v, \sum_{i=1}^{|T|} \lambda_i p_i) \mid (\lambda_i)_{i=1}^{|T|} \in \Delta(\{1, \dots, |T|\}), \forall i, (v, p_i) \in G \right\}$
- $\text{IR} = \{v \in \mathbb{R}^T \mid \forall s \in S, \exists y \in \Delta(A), \forall t \in T, v_t \geq u_S(t, s, y)\}$

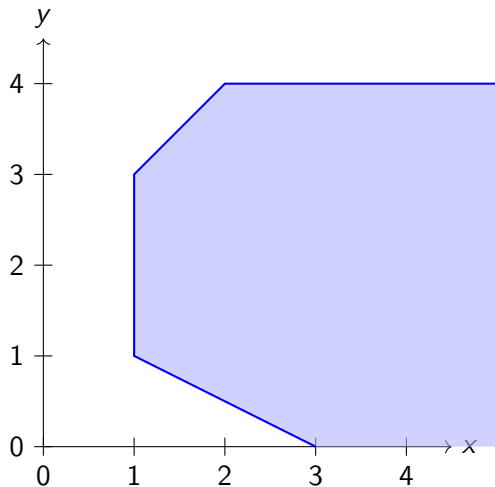
A polyhedron is a set of the form

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}.$$

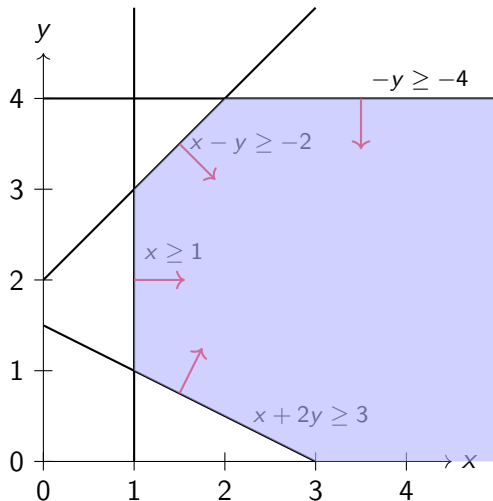
Equivalently, P is the intersection of finitely many half-spaces.

(A, b) is called the H-representation of P .

Polyhedron



Polyhedron



The Minkowski–Weyl Theorem

Theorem

A set $P \subset \mathbb{R}^n$ is a polyhedron iff

$$P = \text{conv}(V) + \text{cone}(R),$$

for some finite sets V, R and where $+$ denotes the Minkowski sum.

R. T. Rockafellar, *Convex Analysis*, 1970, p. 171.

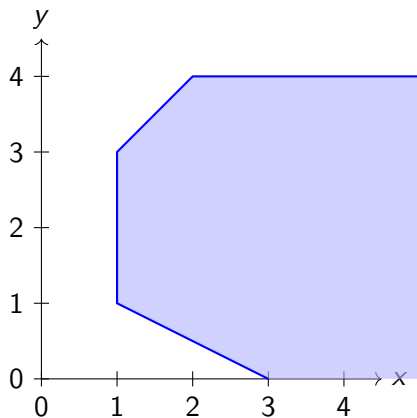
with

- $\text{conv}(V)$: convex combinations (vertices)
- $\text{cone}(R)$: nonnegative combinations (rays)

(V, R) is called the V-representation of P .

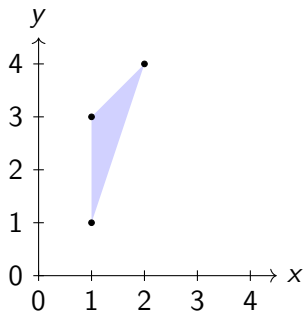
The Minkowski–Weyl Theorem

What is the V-rep of

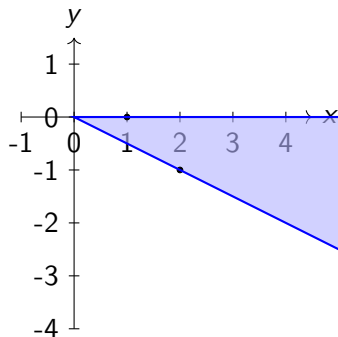


The Minkowski–Weyl Theorem

$$\text{conv}\{(1, 1), (1, 3), (2, 4)\}$$

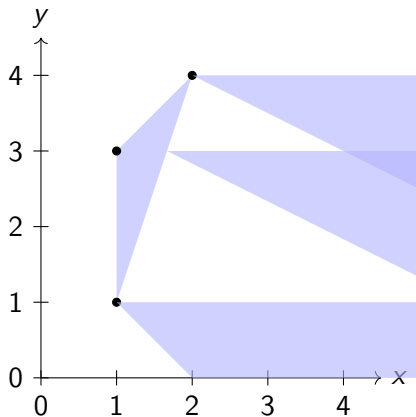


$$\text{cone}\{(1, 0), (2, -1)\}$$



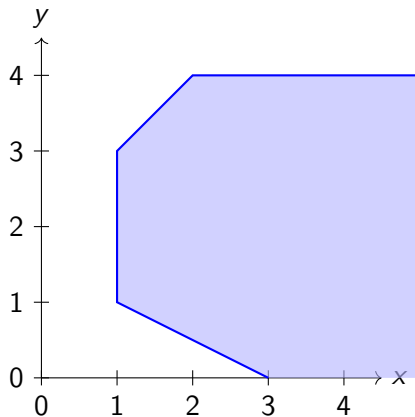
The Minkowski–Weyl Theorem

$$\text{conv}\{(1, 1), (1, 3), (2, 4)\} + \text{cone}\{(1, 0), (2, -1)\}$$



The Minkowski–Weyl Theorem

$$\text{conv}\{(1, 1), (1, 3), (2, 4)\} + \text{cone}\{(1, 0), (2, -1)\}$$



The Minkowski–Weyl Theorem : H-rep vs V-rep

H-rep	V-rep
Allows linear constraints	Is better for representation, you can print the generators
Easier to compute intersection of Polyhedrons	Easier to compute images of linear functions on a Polyhedron
	Easier to compute products of Polyhedrons
	Easier to compute join of Polyhedrons

Some software allows to efficiently go from a representation to another.

H-rep of an intersections of Polyhedrons

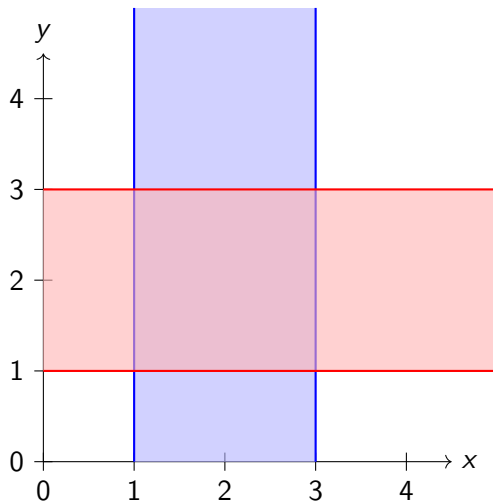
If $(P_i)_I$ a collection of polyhedrons, let (A_i, b_i) be the H-rep of P_i .

Lemma

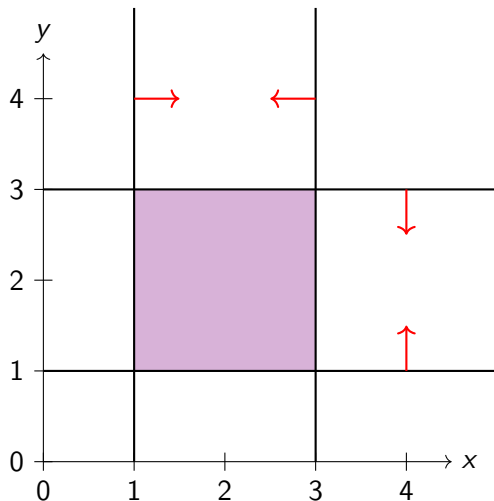
The H-rep of $\bigcap_i P_i$ is

$$\left(\begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix}, \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \right)$$

H-rep of an intersections of Polyhedrons



H-rep of an intersections of Polyhedrons



V-rep of the image of a polyhedron under a linear map

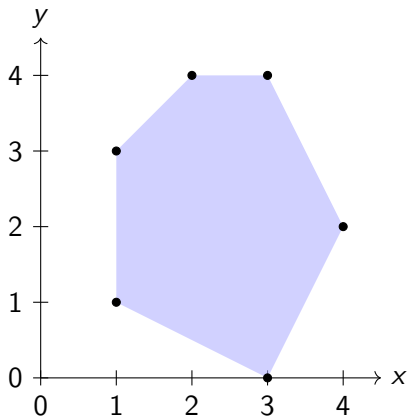
Let f be a linear function and (V, R) the V-rep of P .

Lemma

The V-rep of $f(P)$ is $(f(V), f(R))$.

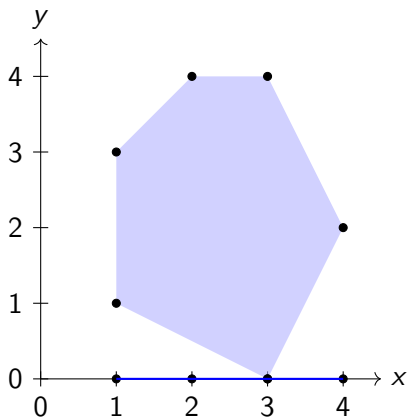
V-rep of the image of a polyhedron under a linear map

Projection on x of



V-rep of the image of a polyhedron under a linear map

Projection on x of



V-rep of a product of Polyhedrons

Let P_1 and P_2 be two polyhedrons with V-rep (V_i, R_i) .

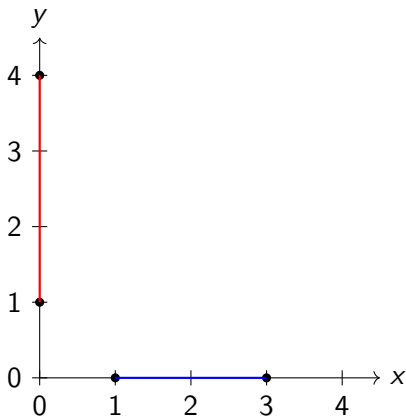
Lemma

The V-rep of $P_1 \times P_2$ is

$$(V_1 \times V_2, (R_1 \times \{0\}) \cup (\{0\} \times R_2))$$

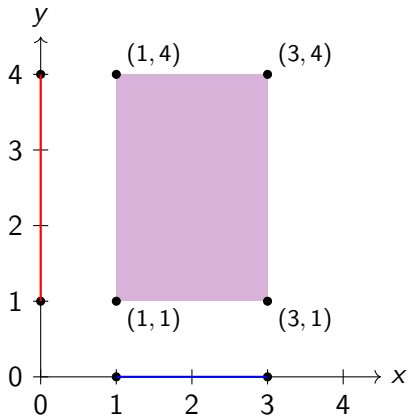
V-rep of a product of Polyhedrons

$$\text{conv}\{1, 3\} \times \text{conv}\{1, 4\}$$



V-rep of a product of Polyhedrons

$$\text{conv}\{1, 3\} \times \text{conv}\{1, 4\}$$



V-rep of a join of Polytopes

Definition (Join)

Let $A_1, \dots, A_n \subset \mathbb{R}^d$. The join of A's is defined as

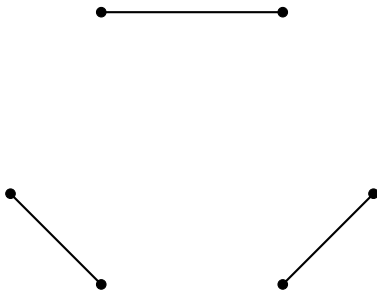
$$A_1 * \dots * A_n = \left\{ \sum_{i=1}^n \lambda_i x_i : x_i \in A_i, \lambda \in \Delta(1, \dots, n) \right\}.$$

Let $(P_i)_I$ be a collection of Polytopes and $(V_i, \emptyset)$ be the V-rep of P_i .

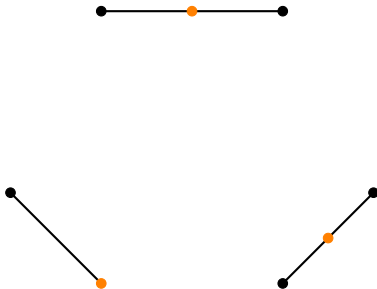
Lemma

*The V-rep of $P_1 * \dots * P_n$ is $(\bigcup_{i \in I} V_i, \emptyset)$.*

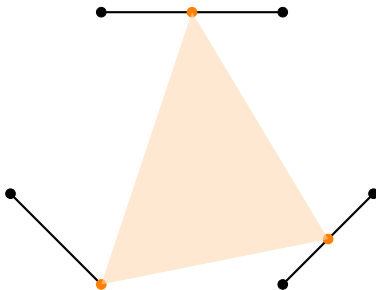
V-rep of a join of Polytopes



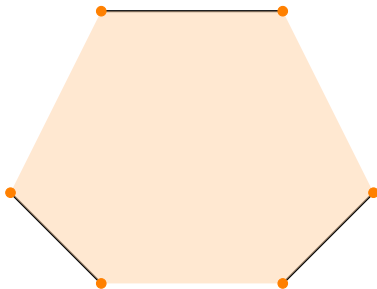
V-rep of a join of Polytopes



V-rep of a join of Polytopes



V-rep of a join of Polytopes



The Minkowski–Weyl Theorem : Application

Let us compute the H-rep of

$$\begin{aligned} C &= \left\{ x \in \mathbb{R}^n \mid \exists y \in \mathbb{R}^m \text{ such that } A \begin{bmatrix} x \\ y \end{bmatrix} \leq b \right\} \\ &= \text{proj}_x \left(\left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^{n+m} \mid A \begin{bmatrix} x \\ y \end{bmatrix} \leq b \right\} \right) \end{aligned}$$

$$\begin{aligned} \text{H-rep of } \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^{n+m} \mid A \begin{bmatrix} x \\ y \end{bmatrix} \leq b \right\} &\longrightarrow \text{V-rep of } \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^{n+m} \mid A \begin{bmatrix} x \\ y \end{bmatrix} \leq b \right\} \\ &\xrightarrow{\text{image of } \text{proj}_x} \text{V-rep of } C \longrightarrow \text{H-rep of } C \end{aligned}$$

For a set E :

- a generic subset of E will be noted $\tilde{E}$,
- The set of non empty subsets of E will be $\mathcal{P}^*(E) = 2^E \setminus \{\emptyset\}$.

2 Computing BNE of signaling games

Recall that $G = \left\{ (v, p) \in \mathbb{R}^{|T|} \times \Delta(T) : \exists s \in S, \exists y \in Y(s, p), \text{ such as } \forall t, v_t \geq u_S(t, s, y) \text{ and } v_t = u_S(t, s, y) \text{ if } p(t) > 0 \right\}$.

Let $K = S \times \mathcal{P}^*(T) \times \mathcal{P}^*(A)$, and for $k = (s, \tilde{T}, \tilde{A}) \in K$,

- $V(k) = \text{proj}_v(\{(v, y) \in \mathbb{R}^T \times \Delta(\tilde{A}) | \forall t, v_t \geq u_S(t, s, y), v_t = u_S(t, s, y) \text{ if } t \in \tilde{T}\})$
- $P(k) = \{p \in \Delta(T) | \Delta(\tilde{A}) \subset Y(s, p), \text{supp}(p) \subset \tilde{T}\}$

Lemma

$$G = \bigcup_{k \in K} V(k) \times P(k)$$

Lemma

$\forall k \in K$, $V(k)$ and $P(k)$ are polyhedrons.

For $k = (s, \tilde{T}, \tilde{A}) \in K$, choose $a_1 \in \tilde{A}$ then

$$V(k) = \text{proj}_v \left\{ (v, y) \left| \begin{array}{l} y(a) \geq 0 \ (a \in \tilde{A}), \quad y(a) = 0 \ (a \notin \tilde{A}), \quad \sum_a y(a) = 1, \\ v_t = \sum_a y(a) U_S(t, s, a) \ (t \in \tilde{T}), \\ v_t \geq \sum_a y(a) U_S(t, s, a) \ (t \notin \tilde{T}) \end{array} \right. \right\}$$

$$P(k) = \left\{ p \left| \begin{array}{l} p(t) \geq 0 \ (t \in T), \quad p(t) = 0 \ (t \notin \tilde{T}), \quad \sum_t p(t) = 1, \\ \sum_t p(t) U_R(t, s, a) = \sum_t p(t) U_R(t, s, a_1) \ (a \in \tilde{A}), \\ \sum_t p(t) U_R(t, s, a) \leq \sum_t p(t) U_R(t, s, a_1) \ (a \notin \tilde{A}) \end{array} \right. \right\}$$

Geometry of $\text{co}_{\text{flat}}(G)$

Recall that, $\text{co}_{\text{flat}}(G) := \left\{ (v, \sum_{i=1}^{|T|} \lambda_i p_i) \mid (\lambda_i)_{i=1}^{|T|} \in \Delta(\{1, \dots, |T|\}), \forall i, (v, p_i) \in G \right\}$.

With, for $\tilde{K} = (k_1, \dots, k_n) \subset K$, let's define

- $V(\tilde{K}) = \bigcap_{k \in \tilde{K}} V(k)$
- $P_{\text{join}}(\tilde{K}) = \ast_{k \in \tilde{K}} P(k)$

Lemma

$$\text{co}_{\text{flat}}(G) = \bigcup_{\substack{\tilde{K} \subset K \\ 1 \leq |\tilde{K}| \leq |T|}} V(\tilde{K}) \times P_{\text{join}}(\tilde{K})$$

Geometry of BNE

Lemma

$$\text{IR} = \bigcap_s \text{proj}_v(\{(v, y) \in \mathbb{R}^T \times \Delta(A) \mid \forall t, v_t \geq u_s(t, s, y)\})$$

Lemma

v is a BNE payoff vector given p iff

$$(v, p) \in \text{BNE} = \bigcup_{\substack{\tilde{K} \subset K \\ 1 \leq |\tilde{K}| \leq |T|}} [V(\tilde{K}) \cap \text{IR}] \times P_{\text{join}}(\tilde{K})$$

Corollary

The set of (v, p) where v is a BNE payoff vector given p , is a finite union of polyhedrons.

An algorithm to compute BNE

The goal is to compute

$$BNE = \bigcup_{\substack{\tilde{K} \subset K \\ 1 \leq |\tilde{K}| \leq |T|}} [V(\tilde{K}) \cap \mathbb{IR}] \times P_{\text{join}}(\tilde{K})$$

Compute the H-representation of IR

Recall that $IR = \bigcap_s \text{proj}_v(\{(v, y) \in \mathbb{R}^T \times \Delta(A) \mid \forall t, v_t \geq u_S(t, s, y)\})$

Algorithm 1 H-representation of IR

```
1 : H_rep_IR  $\leftarrow$  []  
2 : for  $s \in S$  do  
3 :    $VY \leftarrow$  V-rep of  $\{(v, y) \mid \forall a \in A, y(a) \geq 0, \sum_a y(a) = 1,$   
      $\forall t \in T, v_t \geq \sum_a y(a) U_S(t, s, a)\}$   
4 :   Add H-rep of  $\{v, \text{ for } (v, y) \text{ in } VY\}$  to H_rep_IR  
5 : end for  
6 : return H_rep_IR
```

Compute the H-representation of $V(k)$

Recall that $K = S \times \mathcal{P}^*(T) \times \mathcal{P}^*(A)$, and $V(k = (s, \tilde{T}, \tilde{A})) = \text{proj}_V(\{(v, y) \in \mathbb{R}^T \times \Delta(\tilde{A}) \mid \forall t, v_t \geq u_S(t, s, y), v_t = u_S(t, s, y) \text{ if } t \in \tilde{T}\})$

Algorithm 2 H-representation of $V((s, \tilde{T}, \tilde{A}))$

- 1 : $VY \leftarrow$ the V representation of $\{(v, y) \mid \forall a \in \tilde{A} \ y(a) \geq 0, \\ \forall a \notin \tilde{A} \ y(a) = 0, \sum_a y(a) = 1, \\ \forall t \in T \ v_t \geq \sum_a y(a) U_S(t, s, a) \\ \forall t \in \tilde{T} \ v_t = \sum_a y(a) U_S(t, s, a)\}$
 - 2 : **return** H-rep of $\{v, \text{ for } (v, y) \text{ in } VY\}$
-

Compute the V-representation of $V(\tilde{K}) \cap \text{IR}$

Recall that $V(\tilde{K}) \cap \text{IR} = \bigcap_{k \in \tilde{K}} V(k) \cap \text{IR}$

Algorithm 3 V-representation of $V(\tilde{K})$

```
1: H_rep_VK  $\leftarrow$  []  
2: Add the H-representation of IR to H_rep_VK  
3: for  $k \in \tilde{K}$  do  
4:   Add the H-representation of  $V(k)$  to H_rep_VK  
5: end for  
6: return V-rep of H_rep_VK
```

Compute the V-representation of $P_{\text{join}}(\tilde{K})$

Recall that $P_{\text{join}}(\tilde{K}) = \text{conv}(\{p_{k,j} \mid k \in \tilde{K}, 1 \leq j \leq m_k\})$ with $P(k) = \text{conv}(\{p_{k,1}, \dots, p_{k,m_k}\})$.

Algorithm 4 V-representation of P_{join}

```
1 : V_rep_  $P_{\text{join}}$  = []  
2 : for  $k = (s_k, \tilde{T}_k, \tilde{A}_k) \in \tilde{K}$  do  
3 :   Add the V-rep of  $P(k)$  to V_rep_  $P_{\text{join}}$   
4 : end for  
5 : return V_rep_  $P_{\text{join}}$ 
```

$$BNE = \bigcup_{\substack{\tilde{K} \subset K \\ 1 \leq |\tilde{K}| \leq |T|}} [V(\tilde{K}) \cap IR] \times P_{\text{join}}(\tilde{K})$$

Algorithm 5 Compute BNE

```
1: for  $\tilde{K} \subset K$  with  $1 \leq |\tilde{K}| \leq |T|$  do
2:   if  $V(\tilde{K}) \cap IR \neq \emptyset$  and  $\forall k \in \tilde{K}, P(k) \neq \emptyset$  then
3:     Print V-rep of  $V(\tilde{K}) \cap IR$  and V-rep of  $P_{\text{join}}(\tilde{K})$ .
4:   end if
5: end for
```

- Worst case : BNE has $S^T e^{T^2+TA}$ polyhedrons.
- It is possible to adapt this algorithm to get BNE strategies, but it isn't polyhedrons any more.
- It is possible to adapt this algorithm to obtain PBE.
- If you want the BNE without cheap talk messages, select only $\tilde{K}$ where $\forall k \neq k', s_k \neq s_{k'}$.
- Some polyhedrons are redundant, if you find them, you cut the number of polyhedrons.

- The algorithm presented is not the one implemented.
- Computation and rendering optimization become important.
- How to present results if $|T| > 2$?
- A numerical algorithm can make mistakes due to rounding.

3 Computing Communication equilibrium

Communication equilibrium

A communication equilibrium is

- a signaling rule $\mu_1 : T \rightarrow \Delta(S)$,
- an action rule $\mu_2 : T \times S \times S \rightarrow \Delta(A)$,

Such as

- Sender incentive constraint. For all $t \in T$ and all deviations $t' \in T, f \in S^S$,

$$\sum_{s \in S} \sum_{a \in A} \mu_1(s | t) \mu_2(a | t, s, s) U_S(t, s, a) \geq \sum_{s \in S} \sum_{a \in A} \mu_1(s | t') \mu_2(a | t', s, f(s)) U_S(t, f(s), a).$$

- Receiver incentive constraint For all $s \in S$ and all deviations $s' \in S, f \in A^A$,

$$\sum_{t \in T} \sum_{a \in A} p(t) \mu_1(s | t) \mu_2(a | t, s, s) U_R(t, s, a) \geq \sum_{t \in T} \sum_{a \in A} p(t) \mu_1(s | t) \mu_2(a | t, s, s') U_R(t, s, f(a)).$$

For a CE, the outcome is given by

$$\pi_{s,a|t} = \mu_1(s|t)\mu_2(a|t, s, s)$$

We denote all the outcomes of all the CE with prior p as $\mathcal{O}^{CE}(G, p)$.

In general we have

$$\mathcal{O}^{BNE}(G, p) \subset \mathcal{O}^{CE}(G, p)$$

Proposition : $\pi \in \mathcal{O}^{CE}(G, p)$ iff $\exists \nu : \{(t, s_1, s_2) \in T \times S \times S \mid s_1 \neq s_2\} \rightarrow \Delta(A)$,

$$(1) \forall t, t', f, \sum_{s,a} \pi_{s,a|t} U_S(t, s, a) \geq \sum_{s,a} 1_{f(s)=s} \pi_{s,a|t'} U_S(t, s, a)$$

$$+ 1_{f(s) \neq s} \left[\sum_{a'} \pi_{s,a'|t'} \nu(a|t', s, f(s)) U_S(t, f(s), a) \right]$$

$$(2) \forall s, s', g, \sum_{t,a} p(t) \pi_{s,a|t} U_R(t, s, a) \geq 1_{s=s'} \sum_{t,a} p(t) \pi_{s,a|t} U_R(t, s, g(a))$$

$$+ 1_{s \neq s'} \sum_{t,a} p(t) \left[\sum_{a'} \pi_{s,a'|t} \nu(a|t, s, s') U_R(t, s, g(a)) \right]$$

We say that (π, ν) is a CE if it satisfies these conditions.

Definition (Clear punishment)

A signaling game has *clear punishment* if there exists a function

$$P : S \rightarrow A$$

such that

$$\forall t \in T, \forall s \in S, \forall a \in A, \quad U_S(t, s, a) \geq U_S(t, s, P(s)).$$

Clear Punishment

Proposition

The set of CE outcomes for a signaling game with clear punishment and prior p is a polyhedron.

It is defined by the following inequality on π :

- $\forall t, t' \in T, \forall f \in S^S, \sum_{s,a} \pi_{s,a|t} U_S(t, s, a) \geq \sum_{s,a} \pi_{s,a|t'} [1_{f(s)=s} U_S(t, s, a) + 1_{f(s) \neq s} U_S(t, f(s), P(f(s)))]$
- $\forall s, a, a', \sum_t p(t) \pi_{s,a|t} U_R(t, s, a) \geq \sum_t p(t) \pi_{s,a'|t} U_R(t, s, a')$

Lemma

If a signaling game has a clear punish P , (π, ν) is a CE implies (π, ν_P) is a CE. With $\nu_P(P(s')|t, s, s') = 1$ for all $t, s \neq s'$.

Example : Beer-Quiche

Here are the generators of the CE of Beer-Quiche with $p(W) = \frac{1}{4}$,

	BN 1	CE 2	BN 2	CE 4
$\frac{\pi_{B,F W}}{\pi_{Q,F W}} \mid \frac{\pi_{B,C W}}{\pi_{Q,C W}}$	$\frac{0}{0} \mid \frac{1}{0}$	$\frac{0}{0.4} \mid \frac{0}{0.6}$	$\frac{0}{0} \mid \frac{0}{1}$	$\frac{0.2}{0} \mid \frac{0.2}{0.6}$
$\frac{\pi_{B,F S}}{\pi_{Q,F S}} \mid \frac{\pi_{B,C S}}{\pi_{Q,C S}}$	$\frac{0}{0} \mid \frac{1}{0}$	$\frac{0}{0} \mid \frac{0.8}{0.2}$	$\frac{0}{0} \mid \frac{0}{1}$	$\frac{0}{0} \mid \frac{0.8}{0.2}$
$E[U_S]$	2.75	2.65	2.25	2.65
$E[U_R]$	0.75	0.85	0.75	0.8

	CE 5	CE 6	CE 7
$\frac{\pi_{B,F W}}{\pi_{Q,F W}} \mid \frac{\pi_{B,C W}}{\pi_{Q,C W}}$	$\frac{0.25}{0} \mid \frac{0.25}{0.5}$	$\frac{0}{0} \mid \frac{0}{1}$	$\frac{0.5}{0} \mid \frac{0.5}{0}$
$\frac{\pi_{B,F S}}{\pi_{Q,F S}} \mid \frac{\pi_{B,C S}}{\pi_{Q,C S}}$	$\frac{1/12}{0} \mid \frac{0.75}{2/12}$	$\frac{0}{0} \mid \frac{2/3}{1/3}$	$\frac{0}{1/6} \mid \frac{2/3}{1/6}$
$E[U_S]$	2.5	2.75	2.25
$E[U_R]$	0.75	0.75	0.75

What could we do next ?